

Algebra II with Trigonometry

Chapter 5.1

Olympiad Problems (GPT-5.4)

Saved tutoring response with MathJax rendering. Original PDF file:
[4401267_alg-2trig-h-chp-5-1-note-docx.pdf](#)

Problems

1. Find the least positive angle in degrees coterminal with

$$\frac{37\pi}{12} - 845^\circ.$$

2. A sector has perimeter 20 and central angle 2 radians. Find its area.
3. In a circle, the numerical value of the arc length of a sector equals the numerical value of the area of that sector. If the central angle is not 0, find it.
4. A point moves on a circle of radius 6 cm. In the first half of a trip it turns through 150° ; in the second half it turns through $\frac{5\pi}{6}$ radians. Find the total distance traveled.
5. Two angles in standard position satisfy

$$\alpha + \beta = 7\pi, \quad \alpha - \beta = 250^\circ.$$

Find the least positive coterminal angle of α .

6. A wheel of radius r rolls without slipping. After exactly 9 revolutions, the distance traveled equals the area of a sector of a circle of radius r and angle θ radians. Find θ .